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LPR ECPI – Integration Guide

TARGET AUDIENCE:

Project management team, software and hardware development team. Limited to technology partners and internal use only **(PRIVATE AND CONFIDENTIAL)**

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Document Version

Version	Date	By	Description
1.0	2023-10-31	SF Lee	Initial Release
1.1	2023-12-26	SF Lee	Added: - ShowStatus command - TngAbt Scheme Id
1.2	2024-05-29	SF Lee	Added: - Updated error code
1.3	2024-07-16	SF Lee	Added: - EntryDt, VehicleNo in InitTxn - Tid, Mid, BatchNo in TxnStatus
1.4	2024-08-16	SF Lee	Added: - Fare Details in InitTxn - Heartbeat command - QR B scan C response Updated: - Process Flow
1.5	2024-09-18	SF Lee	Added: - entryLane in initTxn - txID in txnStatus response
1.6	2024-09-25	SF Lee	Added: - Fare Details in InitTxn
1.7	2024-10-11	Seow Yen	Added: - Heartbeat - Process flow - Signature Calculation and signed message - Logging between Kiosk Controller and Reader - Appendix B
1.8	2024-10-30	Seow Yen	Correction: - Updated 4.6 and 4.7 for cardScheme
1.9	2024-11-01	Justin	Added SCPI Tunnel command
1.10	2024-11-26	Seow Yen	Added - Set reader IP using TCPI tool Updated

			- Process Flow
1.11	2025-03-03	SF Lee	Added: - Txn Query command
1.12	2025-04-03	SF Lee	Added: - Delay Online mode in InitTerminal - writeEntry and returnEntry in InitCard
1.13	2025-05-15	Justin Aw	Remove Delay Online mode in InitTerminal. Replaced with offlineApprove in initTerminal command Correction: - Scpi response type corrected to "response"
1.14	2025-07-31	Justin Aw	Added SoundID in AbortTxn
1.15	2025-08-11	Justin Aw	Corrected message and title to messageTXT and titleTXT in initTerminal and initCard command.
1.16	2025-10-14	Justin Aw	Added Get IO Value, and Set IO Value
1.17	2025-10-29	Justin Aw	Correction on Get IO Value, and Set IO Value Added: initCard, added all response data in message, and to return TNG entry information if returnEntry flag is set.

Table of Contents

1 Introduction

1.1 Scope

This document outlines the integration API between reader and LPR controller:

1. Overview, Connectivity & Setup
2. Message format
3. Commands & Responses
4. Process Flow

1.2 Abbreviations

EMV	Europay, Mastercard, Visa
Jellies	CoherentPlus Contactless Reader
ODA	Offline Data Authentication
LPR	License Plate Recognition

1.3 References

NA

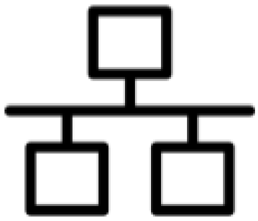
2 Overview, Connectivity & Setup

2.1 Overview

ECPI is a communication interface that facilitates payment functionality for LPR to accept EMV card, local electronics purse payment and E-Wallet on parking solution. LPR and Jellies readers are connected via LAN communication. This document describes the command to operate Jellies to accept payment.

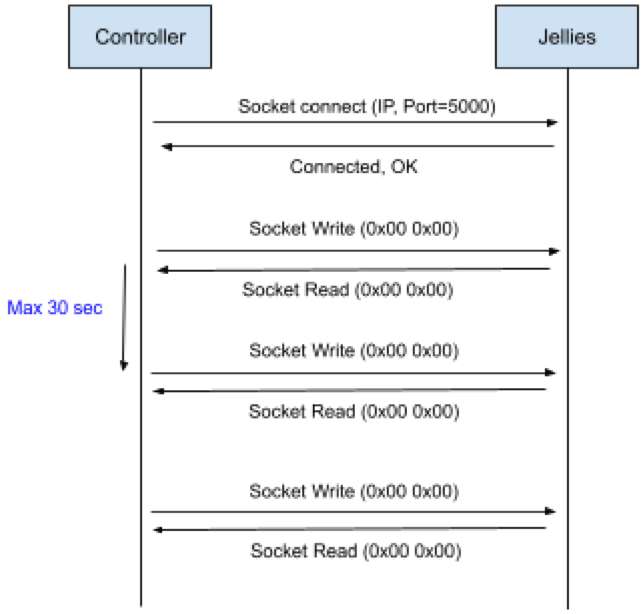
The commands are designed for simple integration, without being required to integrate the bank system. Jellies will handle the EMV, local electronics purse card reading, E-Wallet and bank host authorization process, and returning LPR a payment status with approval detail.

2.2 Connectivity & Setup



A. Socket (TCP via LAN)
   a. Default socket is opened at Port 5000
   b. Connect to this device IP with port number
   c. One connection allowed per session. The second connection will be refused if the first connection is still alive.

B. Heartbeat
   a. Heartbeat is required to send in every 30 seconds (within 30 seconds) if there is no communication in between.
   b. Connection will be dropped by the reader if there is no heartbeat communicated within the heartbeat cutoff period. When the connection is dropped, the controller is required to recover the connection by reestablishing the connection and initialization process as per illustrated in Process Flow.
   c. Heartbeat is a 2 bytes message carrying null bytes, 0x00 0x00.



3 Message Format

This chapter describes the general format of requests and responses that communicate between reader and LPR. The supported message format for all the requests and responses is in JSON format.

3.1 Request

Key Name	Max Length	Description
apiVersion	12	Api Version.
message	20	Command of this message.
type	10	Type of this message. request - Request
timestamp	20	Date time of this message. 2023-05-30 11:06:02 - YY-MM-DD hh:mm:ss
messageTraceId	40	Trace Id for this message.
body	n	Body of the message.
signature	64	SHA256 of the full message with secret key. signature = secretkey + full message


```
{
        "apiVersion":"1.0",
        "message":"initTerminal",
        "type":"request",
        "timestamp":"2023-05-30 11:06:02",
        "messageTraceID":"112233445566",
        "body":{
                "plazaID":"P01",
                "laneID":"L01",
                "operationMode":"1",
                "laneType":"3"
        },
        "signature":"0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706"
}
```

3.2 Response

<table>
<tr><th>Key Name</th><th>Max Length</th><th colspan="2">Description</th></tr>
<tr><td>apiVersion</td><td>12</td><td colspan="2">Api Version.</td></tr>
<tr><td>message</td><td>20</td><td colspan="2">Command of this message.</td></tr>
<tr><td>type</td><td>10</td><td colspan="2">Type of this message.

ack - Acknowledgement</td></tr>
<tr><td>timestamp</td><td>20</td><td colspan="2">Date time of this message.

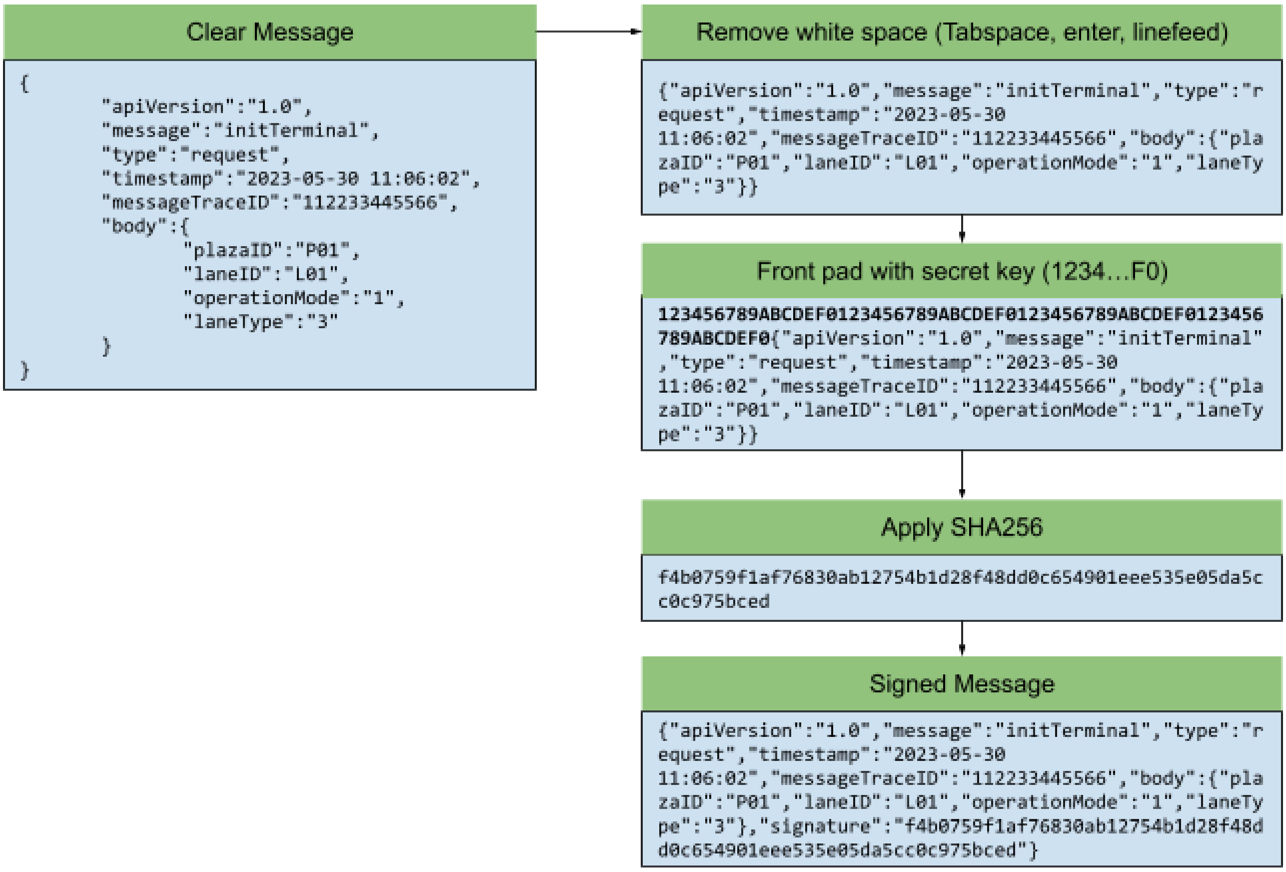
2023-05-30 11:06:02 - YY-MM-DD hh:mm:ss</td></tr>
<tr><td>messageTraceId</td><td>40</td><td colspan="2">Trace Id for this message.</td></tr>
<tr><td rowspan="2">body</td><td rowspan="2">n</td><td colspan="2">Body of the message.</td></tr>
<tr><td>errorCode[4]</td><td>Response status of command execution</td></tr>
<tr><td>signature</td><td>64</td><td colspan="2">SHA256 of the full message with secret key.

signature = secretkey + full message</td></tr>
</table>


```
{
	"apiVersion":"1.0",
	"message":"initTerminal",
	"type":"ack",
	"timestamp":"2023-05-30 11:06:02",
	"messageTraceID":"112233445566",
	"body":{
		"errorCode":"0000"
	},
	"signature":"ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA"
}
```

3.3 Signature Calculation and Signed Message

The flow diagram below illustrates the signature calculation and generating a signed message.



3.4 Logging between Kiosk Controller and Reader

System integrators are required to log all communications between the controller and the reader. This should include timestamps, IP addresses, network connection activities, socket messages, and any exceptions that occur during message exchanges, as shown below:

A. A normal scenario:

```
2024-05-23 10:05:10 [192.168.0.50] >> Connect Port 500
2024-05-23 10:05:10 [192.168.0.50] << Connected Successfully
2024-05-23 10:05:11 [192.168.0.50] >> SENT "{abcdefg………………….."}
2024-05-23 10:05:12 [192.168.0.50] << RCVD "{………………{"errorCode = 0000"}……………."}
2024-05-23 10:05:13 [192.168.0.50] << RCVD "{………………{"errorCode = 0000"}……………."}
2024-05-23 10:05:14 [192.168.0.50] >> SENT "{abcdefg………………….."}
2024-05-23 10:05:15 [192.168.0.50] << RCVD "{………………{"errorCode = 0000"}……………."}
2024-05-23 10:05:14 [192.168.0.50] >> SENT "{abcdefg………………….."}
2024-05-23 10:05:15 [192.168.0.50] << RCVD "{………………{"errorCode = 0000"}……………."}
2024-05-23 10:05:16 [192.168.0.50] >> SENT "{abcdefg………………….."}
2024-05-23 10:05:17 [192.168.0.50] << RCVD "{………………{"errorCode = 0000"}……………."}
2024-05-23 10:05:18 [192.168.0.50] >> Disconnect 500
2024-05-23 10:05:19 [192.168.0.50] << Disconnected Successfully
```

B. Example logs of exception errors.

```
2024-05-23 10:05:10 [192.168.0.50] >> Connect Port 500
2024-05-23 10:05:10 [192.168.0.50] << ERROR: Failed to connect to port 500 (Conn refused)

2024-05-23 10:05:11 [192.168.0.50] >> SENT "{abcdefg…………………."}
2024-05-23 10:05:15 [192.168.0.50] << ERROR: Failed to write on socket.

2024-05-23 10:05:11 [192.168.0.50] >> SENT "{abcdefg…………………."}
2024-05-23 10:05:15 [192.168.0.50] << ERROR: Timeout. Unable to receive response.

2024-05-23 10:09:16 [192.168.0.50] << ERROR: Connection terminated.

2024-05-23 10:15:18 [192.168.0.50] << ERROR: Connection closed by remote device
```

This log is crucial for debugging and issue analysis between the system integrator and CoherentPlus. In the event of exceptional cases during development or in a live environment, the log should be submitted to CoherentPlus for initial troubleshooting and resolution.

It is mandatory to retain this log in the controller for at least 15 days, and doing so is highly advisable for effective issue tracking.

4 Commands & Responses

This section outlines the command set and its response respectively. A valid command message shall be packed with message format as described in the previous chapter.

4.1 Init Terminal

The reader will always be in "not-in-use" mode after start up. This command activates the reader into "ready to use" state. LPR is required to pass in the plazaId, laneId, operationMode, laneType. The timestamp will be used to sync the date time to the reader.

Request

<table>
<tr><th>Key Name</th><th>Max Length</th><th colspan="2">Value</th></tr>
<tr><td>apiVersion</td><td>12</td><td colspan="2">1.0</td></tr>
<tr><td>message</td><td>20</td><td colspan="2">initTerminal</td></tr>
<tr><td>type</td><td>10</td><td colspan="2">request</td></tr>
<tr><td>timestamp</td><td>20</td><td colspan="2">2023-05-30 11:06:02</td></tr>
<tr><td>messageTraceID</td><td>40</td><td colspan="2">112233445566</td></tr>
<tr><td rowspan="10">body</td><td rowspan="10">n</td><td>plazaID[10]</td><td>P01</td></tr>
<tr><td>laneID[10]</td><td>L01</td></tr>
<tr><td>operationMode[2]</td><td>0 - Maintenance Mode
1 - Live Mode
2 - Not In Use</td></tr>
<tr><td>laneType[2]</td><td>1 - Entry
2 - Exit
3 - Open
4 - Dual</td></tr>
<tr><td colspan="2">Optional Fields</td></tr>
<tr><td>titleTXT[18]</td><td>Optional. Title text to display
(ecpi-lpr V6.10.2 above)</td></tr>
<tr><td>messageTXT[40]</td><td>Optional. Message text to display
(ecpi-lpr V6.10.2 above)</td></tr>
<tr><td>selectNext[2]</td><td>Optional. To allow unknown card to be processed by controller on APDU level
0 - Unknown card will be alert with error message
1 - Unknown card will NOT be alerted with an error message, but return invalid card error to the controller to process with other card business rules on APDU level, and process the alert on controller level.</td></tr>
<tr><td>offlineApprove[2]</td><td>Optional: To allow payment terminal to approve EMV transaction in offline condition below:</td></tr>
</table>

			0 - Always online. Connection/Timeout error will end with declined. 1 - Auto Offline. If connection/Timeout detected, transaction will be stored in batch file for delay online approval. 2- Always Offline. Always approved offline, transactions will be stored in batch files for delay online approval. Delay online authorization is scheduled and automated. **This configuration supported from ECPI-LPR V6.9.3** **NOTE: Enable offlineApprove (Delay authorization) is on MERCHANT RISK. There is no guarantee that delayed authorization will be approved.**
signature	64	0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706	


```
{
        "apiVersion":"1.0",
        "message":"initTerminal",
        "type":"request",
        "timestamp":"2023-05-30 11:06:02",
        "messageTraceID":"112233445566",
        "body":{
                "plazaID":"P01",
                "laneID":"L01",
                "operationMode":"1",
                "laneType":"4"
        },
        "signature":"0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706"
}
```


<u>Response</u>

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initTerminal	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	aa11bb22cc33dd44	
body	n	errorCode[4]	0000
signature	64	ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA	


```
{
        "apiVersion":"1.0",
        "message":"initTerminal",
        "type":"ack",
        "timestamp":"2023-05-30 11:06:02",
        "messageTraceID":"112233445566",
        "body":{
                "errorCode":"0000"
        },
        "signature":"ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA"
```

}

4.2 Get Status

This function is to get the current reader status and the previous response. LPR is required to send this command to get the previous response if the current command is timeout.

Request

Key Name	Max Length	Value
apiVersion	12	1.0
message	20	getStatus
type	10	request
timestamp	20	2023-05-30 11:06:02
messageTraceID	40	112233445566
body	n	empty
signature	64	48E412867FA461765876EF0903233E857823D28BB52169E7DE1EC1C3CB3CFBCB


```
{
	"apiVersion":"1.0",
	"message":"getStatus",
	"type":"request",
	"timestamp":"2023-05-30 11:06:02",
	"messageTraceID":"112233445566",
	"body":{},
	"signature":"48E412867FA461765876EF0903233E857823D28BB52169E7DE1EC1C3CB3CFBCB"
}
```

Response

<table>
<tr><th>Key Name</th><th>Max Length</th><th colspan="2">Value</th></tr>
<tr><td>apiVersion</td><td>12</td><td colspan="2">1.0</td></tr>
<tr><td>message</td><td>20</td><td colspan="2">getStatus</td></tr>
<tr><td>type</td><td>10</td><td colspan="2">ack</td></tr>
<tr><td>timestamp</td><td>20</td><td colspan="2">2023-05-30 11:06:02</td></tr>
<tr><td>messageTraceID</td><td>40</td><td colspan="2">112233445566</td></tr>
<tr><td rowspan="5">body</td><td rowspan="5">n</td><td>errorCode</td><td>0000</td></tr>
<tr><td>operationMode</td><td>0 - Maintenance Mode
1 - Live Mode
2 - Not In Use</td></tr>
<tr><td>state</td><td>01 - Idle
02 - Scanning Card
03 - Card Detecting
04 - Card Detected
05 - Host Authorizing
03 - Transaction Completed
99 - Not In Use</td></tr>
<tr><td>lastCmd</td><td>Full command from previous instruction</td></tr>
<tr><td>lastResp</td><td>Full response from previous instruction</td></tr>
<tr><td>signature</td><td>64</td><td colspan="2">BE6C458580F8C6256E72D4FBE62C5F6A42BD37470D82273374285B541DDC124F</td></tr>
</table>


```
{
        "apiVersion":"1.0",
        "message":"getStatus",
        "type":"ack",
        "timestamp":"2023-05-30 11:11:21",
        "messageTraceID":"112233445588",
        "body":{
                "errorCode":"0000",
                "operationMode:0",
                "state":"1",
                "lastCmd":{
                  "apiVersion":"1.0",
                  "message":"initTerminal",
                  "type":"request",
                  "timestamp":"2023-05-30 11:06:02",
                  "messageTraceID":"112233445566",
                  "body":{
                          "plazaID":"P01","laneID":"L01","operationMode":"1","laneType":"3"
                  },
                  "signature":"0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706"
                },
                "lastResp":{
                  "apiVersion":"1.0",
                  "message":"initTerminal",
                  "type":"request",
                  "timestamp":"2023-05-30 11:06:02",
                  "messageTraceID":"112233445566",
                  "body":{
                          "errorCode":"0000"
                  },
```

```
            "signature":"ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA"
        }
    },
    "signature":"BE6C458580F8C6256E72D4FBE62C5F6A42BD37470D82273374285B541DDC124F"
}
```

4.3 Init Card

This function is to start the process of detecting and reading a card. If the card is found valid, it will send a Card Read command to LPR. LPR is required to send this command on every end of the transaction if retrigger is present and set to 1, otherwise the reader will go back to detecting and reading a card after Finish Txn. This function is optional.

For entry, it is optional to send InitTxn if it is an offline entry with ODA process. LPR is required to send Finish Txn to complete the process without charging the card.

For exit, it is required to send InitTxn if it is charging the card.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initCard	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	fareClass[2]	This field to be defined by LPR
		retrigger[1]	0 - Do not required to send initCard. Reader will start scanning card after finTxn. 1 - Required to send initCard to start scanning card.
		writeEntry[1]	Optional. To write entry info for TNG card 0 - Do not require to write entry info for TNG card. 1 - Required to write entry info for TNG card
		returnEntry[1]	Optional. To return entry info for TNG card 0 - Do not require to return entry info for TNG card. 1- Required to return entry info for TNG card.
		titleTXT[18]	Optional. Title text to display *(ecpi-lpr V6.10.2 above)*
		messageTXT[40]	Optional. Message text to display *(ecpi-lpr V6.10.2 above)*
signature	64	0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706	


```
{
        "apiVersion":"1.0",
        "message":"initCard",
        "type":"request",
        "timestamp":"2023-05-30 11:06:02",
        "messageTraceID":"112233445566",
        "body":{
                "fareClass":"P01",
                "retrigger":"1",
```

```
        },
        "signature":"0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706"
}
```


Response

<table>
<tr><th>Key Name</th><th>Max Length</th><th colspan="2">Value</th></tr>
<tr><td>apiVersion</td><td>12</td><td colspan="2">1.0</td></tr>
<tr><td>message</td><td>20</td><td colspan="2">initCard</td></tr>
<tr><td>type</td><td>10</td><td colspan="2">ack</td></tr>
<tr><td>timestamp</td><td>20</td><td colspan="2">2023-05-30 11:06:02</td></tr>
<tr><td>messageTraceID</td><td>40</td><td colspan="2">112233445566</td></tr>
<tr><td rowspan="11">body</td><td rowspan="11">n</td><td>errorCode[4]</td><td>0000</td></tr>
<tr><td>txnID</td><td>Terminal transaction ID. Generated from terminal.

0A6CDF1690177983701</td></tr>
<tr><td>txnDt[20]</td><td>2023-05-30 11:06:02</td></tr>
<tr><td>maskPan[20]</td><td>Masked PAN. Only first 6 and last 4 will be returned in clear, others will be masked with *. Please refer to Appendix B.

4935 31** **** 2683</td></tr>
<tr><td>hashPan[40]</td><td>Hashed PAN. Hash with a full clear Card PAN.

649B28A9F095B9716AA9543067ED4F9A995E8F62</td></tr>
<tr><td>cardScheme[2]</td><td>01 - MyDebit
02 - Mastercard
03 - Visa
11 - Tng
14 - TngAbt</td></tr>
<tr><td>cardBalance[10]</td><td>The balance of TNG card (If cardScheme = Tng)</td></tr>
<tr><td colspan="2">The following additional TNG data returned if “returnEntry” in Request set to 1.</td></tr>
<tr><td>entryDt[20]</td><td>Entry Date Time
2023-05-30 11:06:02</td></tr>
<tr><td>entryOpId[2]</td><td>Entry TNG Operator ID</td></tr>
<tr><td>entryLocId[3]</td><td>Entry TNG LocationID</td></tr>
</table>

		entryLaneId[2]	Entry TNG LaneID
signature	64	ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA	


```
{
        "apiVersion":"1.0",
        "message":"initCard",
        "type":"ack",
        "timestamp":"2023-05-30 11:06:02",
        "messageTraceID":"112233445566",
        "body":{
                "errorCode":"0000"
        },
        "signature":"ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA"
}
```

4.4 Init Transaction

This function is to start the process of card scanning for LPR exit. This function read card, validates internally (i.e blacklist, whitelist, expiry check). If the card is found valid and authorization to acquiring bank, it will send a Txn Status command to LPR. LPR is required to send this command on every end of the transaction if retrigger is present and set to 1, otherwise the reader will go back to card scanning after Finish Txn.

<u>Request</u>

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initTxn	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	txnType[12]	Optional. Transaction type of this transaction to differentiate Entry or Exit.
		fareAmount[12]	350
		fareClass[2]	This field to be defined by LPR
		retrigger[1]	0 - Do not required to send initTxn. Reader will start scanning card after finTxn. 1 - Required to send initTxn to start scanning card.
		entryDt[20]	Entry Date Time for exit transaction 2023-05-30 11:06:02
		vehicleNo[10]	Vehicle Plate No VMM1234
		entryLane[10]	Entry Lane E01
		gstAmount	Gst Amount 0
		gstTaxAmount	Gst TaxAmount 0
		sAmount	Service Charge Amount 0

		sTaxAmount	Server Charge Tax Amount 0
		pAmount	Parking Amount 350
		pTaxAmount	Parking Tax Amount 0
		discAmount	Discount Amount 0
		discTaxAmount	Discount Tax Amount 0
		title[18]	Optional. Title text to display
		message[40]	Optional. Message text to display
signature	64	1BE1581D1CA63EBD65D919DFFB7F575DDC396A5C4822CE6F395C1BEC34010E39	


```
{
        "apiVersion":"1.0",
        "message":"initTxn",
        "type":"request",
        "timestamp":"2023-05-30 11:06:02",
        "messageTraceID":"112233445566",
        "body":{
                "fareAmount":"350",
                "fareClass":"1",
                "retrigger":"1",
                "entryDt":"2023-05-30 11:06:02",
                “vehicleNo”:”VMM1234”,
                “entryLane”:”E01”,
                “gstAmount”:”0”,
                “gstTaxAmount”:”0”;
                “sAmount”:”0”,
                “sTaxAmount”:”0”,
                “pAmount”:”350”,
                “pTaxAmount”:”0”,
                “discAmount”:”0”,
                “discTaxAmount”:”0”,
        },
        "signature":"1BE1581D1CA63EBD65D919DFFB7F575DDC396A5C4822CE6F395C1BEC34010E39"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	initTxn	
type	10	ack	
timestamp	20	2023-05-30 11:07:32	
messageTraceID	40	112233445566	
body	n	errorCode	0000
signature	64	29DD4219279A020368A102BBF57A2EBA958BC43D8EC7A4F8EB04E9C063A6D075	


```
{
        "apiVersion":"1.0",
        "message":"initTxn",
        "type":"ack",
        "timestamp":"2023-05-30 11:07:32",
        "messageTraceID":"112233445566",
        "body":{
                "errorCode":"0000"
        },
        "signature":"29DD4219279A020368A102BBF57A2EBA958BC43D8EC7A4F8EB04E9C063A6D075"
}
```

4.5 Abort Txn

This function is to abort the card scanning process and return to idle state.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	abortTxn	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	titleTXT[18]	Optional. Title text to display
		messageTXT[40]	Optional. Message text to display
		soundID[2]	*(jvx-ecpi-lpr version 6.9.11 onward)* Optional: Sound Id - 01 (Success beep) - 02 (Failed beep) - FF (No beep) - Default no beep
signature	64	4083E507079483AB35B37794D366E0EFE6BFC1647E10FA9FCA1CBE3C8E200DD1	


```
{
	"apiVersion":"1.0",
	"message":"abortTxn",
	"type":"request",
	"timestamp":"2023-05-30 11:06:02",
	"messageTraceID":"112233445566",
	"body":{},
	"signature":"4083E507079483AB35B37794D366E0EFE6BFC1647E10FA9FCA1CBE3C8E200DD1"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	abortTxn	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
signature	64	79C591841928BC2010E247729C6873C30206BA2B50F26261C4588E300F71DF02	


```
{
	"apiVersion":"1.0",
	"message":"abortTxn",
	"type":"ack",
	"timestamp":"2023-05-30 11:22:54",
	"messageTraceID":"112233445566",
	"body":{
		"errorCode":"0000"
	},
	"signature":"79C591841928BC2010E247729C6873C30206BA2B50F26261C4588E300F71DF02"
}
```

4.6 Card Read

This function is sent from the reader to LPR when card is detected and the status of the card for Init Card (i.e valid card, blacklisted, expired card).

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	cardRead	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445577	
body	n	errorCode[4]	0000
		txnID	Terminal transaction ID. Generated from terminal. 0A6CDF1690177983701
		txnDt[20]	2023-05-30 11:06:02
		maskPan[20]	Masked PAN. Only first 6 and last 4 will be returned in clear, others will be masked with *. Please refer to Appendix B. 4935 31** **** 2683
		hashPan[40]	Hashed PAN. Hash with a full clear Card PAN. 649B28A9F095B9716AA9543067ED4F9A995E8F62
		cardScheme[2]	01 - MyDebit 02 - Mastercard 03 - Visa 04 - Amex 11 - Tng 12 - Ezlink 13 - Nets 14 - TngAbt
signature	64	5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97	


```
{
        "apiVersion":"1.0",
        "message":"cardRead",
        "type":"request",
        "timestamp":"2023-05-30 11:06:02",
        "messageTraceID":"112233445577",
        "body":{
                "errorCode":"0000",
                "txnDt":"2023-05-30 11:06:02",
                "maskPan":"493531******2683",
```

```
            "hashPan":"649B28A9F095B9716AA9543067ED4F9A995E8F62",
            "cardScheme":"01"
      },

      "signature":"5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	cardRead	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445577	
body	n	errorCode	0000
signature	64		


```
{
	"apiVersion":"1.0",
	"message":"cardRead",
	"type":"ack",
	"timestamp":"2023-05-30 11:06:02",
	"messageTraceID":"112233445577",
	"body":{
		"errorCode":"0000"
	},
	"signature":"5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97"
}
```

4.7 Txn Status

This function is sent from the reader to LPR on the status of the transaction for Init Txn (i.e approved, declined from the acquiring host).

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	txnStatus	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445577	
body	n	errorCode[4]	0000
		txnID	Terminal transaction ID. Generated from terminal. 0A6CDF1690177983701
		paymentMode	Transaction Payment Mode. online/offline
		txnDt[20]	2023-05-30 11:06:02
		maskPan[20]	Masked PAN. Only first 6 and last 4 will be returned in clear, others will be masked with *. Please refer to Appendix B. 4935 31** **** 2683
		hashPan[40]	Hashed PAN. Hash with a full clear Card PAN. 649B28A9F095B9716AA9543067ED4F9A995E8F62
		cardScheme[2]	01 - MyDebit 02 - Mastercard 03 - Visa 11 - Tng 14 - TngAbt
		EMV Card	
		txID	Transaction ID. Generated from back office 172665812756161008
		mid[15]	Merchant ID from acquirer bank
		tid[8]	Terminal ID from acquirer bank

		stan[6]	System Trace Number. Sent from reader to acquiring host. 000001
		rrn[12]	Retrieval Reference Number. Returned from acquiring host. 000012300001
		apprCode[6]	Approval Code. Returned from acquiring host. 123456
		invNo[6]	Invoice Number. Sent from reader to acquiring host. 000001
		batchNo[6]	Batch No
		TNG Card	
		cardBalance	Card Balance 1667
		cardCsc	Card Txn Counter 63885
		nrtPayload	Payload for NRT Server (*Controller is required to store this payload in case the reader breaks down and is unable to upload the batch)
		QR - B Scan C	
		walletId	E-Wallet ID S92 - TNGD S93 - BOOST S94 - GRABPAY S95 - MAE S96 - SPAY S97 - ALIPAY S98 - WECHATPAY
		fareAmount	Fare Amount. 100
		fareClass	Fare Class. 1
signature	64	5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97	

```
{
	"apiVersion":"1.0",
	"message":"txnStatus",
	"type":"request",
	"timestamp":"2023-05-30 11:06:02",
	"messageTraceID":"112233445577",
	"body":{
		"errorCode":"0000",
		“txnID”:”0A6CDF1690177983701”,
		“paymentMode”:”online”,
		"txnDt":"2023-05-30 11:06:02",
		"maskPan":"493531******2683",
		"hashPan":"649B28A9F095B9716AA9543067ED4F9A995E8F62",
		"cardScheme":"01",
		“txID”:”172665812756161008”,
		“mid”:”12345678”,
		“tid”:”87654321”,
		"stan":"000001",
		"rrn":"000012300001",
		"apprCode":"123456",
		"invNo":"000001",
		“batchNo”:”654321”;
		"fareAmount":"100",
		"fareClass":"1"
	},

	"signature":"5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97"
}
```

<u>Response</u>

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	txnStatus	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445577	
body	n	errorCode	0000
signature	64		


```
{
        "apiVersion":"1.0",
        "message":"txnStatus",
        "type":"ack",
        "timestamp":"2023-05-30 11:06:02",
        "messageTraceID":"112233445577",
        "body":{
                "errorCode":"0000"
        },
        "signature":"5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97"
}
```

4.8 Txn Query

This function is to query the offline transaction posting status. Only valid offline transactions in the batch will be returned.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	txnQuery	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	txnID[18]	Terminal transaction ID. Return from txnStatus.
signature	64	4083E507079483AB35B37794D366E0EFE6BFC1647E10FA9FCA1CBE3C8E200DD1	


```
{
        "apiVersion":"1.0",
        "message":"txnQuery",
        "type":"request",
        "timestamp":"2023-05-30 11:06:02",
        "messageTraceID":"112233445566",
        "body":{
                "txnID":"0A6CDF1690177983701"
        },
        "signature":"4083E507079483AB35B37794D366E0EFE6BFC1647E10FA9FCA1CBE3C8E200DD1"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	txnQuery	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
		txnID	Terminal transaction ID. Send from kiosk
		txnStatus	Transaction Status Transaction approved - approved Transaction rejected - rejected Transaction pending upload - pending Transaction not found - none
signature	64	79C591841928BC2010E247729C6873C30206BA2B50F26261C4588E300F71DF02	


```
{
        "apiVersion":"1.0",
        "message":"txnQuery",
        "type":"ack",
        "timestamp":"2023-05-30 11:22:54",
        "messageTraceID":"112233445566",
        "body":{
                "errorCode":"0000",
                “txnID”:”173349912994442012”,
                "txnStatus":"approved"
        },
        "signature":"79C591841928BC2010E247729C6873C30206BA2B50F26261C4588E300F71DF02"
}
```

4.9 Finish Transaction

This function is to complete the transaction. TCS sends this command after the barrier is opened and the reader will refresh the screen. If this command is not received by the reader, the reader screen will remain at the Transaction Completed screen for operation to support.

Request

Key Name	Max Length	Value
apiVersion	12	1.0
message	20	finTxn
type	10	request
timestamp	20	2023-05-30 11:06:02
messageTraceID	40	112233445566
body	n	
signature	64	9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442


```
{
	"apiVersion":"1.0",
	"message":"finTxn",
	"type":"request",
	"timestamp":"2023-05-30 11:06:02",
	"messageTraceID":"112233445566",
	"body":{},
	"signature":"9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	finTxn	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
signature	64	228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD	


```
{
	"apiVersion":"1.0",
	"message":"finTxn",
	"type":"ack",
	"timestamp":"2023-05-30 11:06:04",
	"messageTraceID":"112233445566",
	"body":{
		"errorCode":"0000"
	},
	"signature":"228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD"
}
```

4.10 Show Status

This function is to show status to the reader. LPR sends this command when the validate on exit is completed without charge and the reader will refresh the screen and reset the state.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	showStatus	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	titleTXT[18]	Title text to display
		messageTXT[40]	Message text to display
		soundId[2]	Sound Id - 01 (Success beep) - 02 (Failed beep) - FF (No beep)
		imageId[2]	Image Id - 04 (Success image) - 08 (Failed image)
signature	64	9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442	


```
{
        "apiVersion":"1.0",
        "message":"showStatus",
        "type":"request",
        "timestamp":"2023-05-30 11:06:02",
        "messageTraceID":"112233445566",
        "body":{
                "titleTXT":"Completed",
                "messageTXT":"FOC",
                "soundID":"01",
                "imageID":"04"
                },
        "signature":"9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442"
}
```

<u>Response</u>

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	showStatus	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
signature	64	228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD	


```
{
        "apiVersion":"1.0",
        "message":"showStatus",
        "type":"ack",
        "timestamp":"2023-05-30 11:06:04",
        "messageTraceID":"112233445566",
        "body":{
                "errorCode":"0000"
        },
        "signature":"228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD"
}
```

4.11 SCPI Tunnel

This function is used to access readers on hardware component level for example NFC PCD, SAM, Buzzer, and etc. Please refer to the SCPI specification to the command list. (Platform and environment functions as well as some functions from SCPI specification may not be supported on this tunnel).

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	scpi	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	sCha[2]	Channel Byte as per SCPI specification
		sIns[2]	Instruction Byte as per SCPI specification
		sData[n]	Data Byte(s) as per SCPI specification
signature	64	9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442	


```
{
        "apiVersion":"1.0",
        "message":"scpi",
        "type":"request",
        "timestamp":"2023-05-30 11:06:02",
        "messageTraceID":"112233445566",
        "body":{
                "sCha":"02",
                "sIns":"12",
                "sData":"010203040506"
                },
        "signature":"9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	scpi	
type	10	response	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
		sStatus[2]	STATUS byte from SCPI response as per SCPI specification
		sRespData[n]	Response DATA as per SCPI specification
signature	64	228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD	


```
{
        "apiVersion":"1.0",
        "message":"scpi",
        "type":"response",
        "timestamp":"2023-05-30 11:06:04",
        "messageTraceID":"112233445566",
        "body":{
                "errorCode":"0000",
                "sStatus":"00",
                "sRespData":"010203040506"
        },
        "signature":"228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD"
}
```

4.12 Set Reader Ip

This function is to configure the reader ip.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	setReaderIp	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	ip	192.168.1.199
		netmask	255.255.255.0
		gateway	192.168.1.1
		dhcp	0
signature	64	9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442	


```
{
	"apiVersion":"1.0",
	"message":"setReaderIp",
	"type":"request",
	"timestamp":"2023-06-26 11:30:30",
	"messageTraceID":"112233445566",
	"body":{
		"ip":"192.168.1.199",
		"netmask":"255.255.255.0",
		"gateway":"192.168.1.1",
		"dhcp":"0"
	},
	"signature":"E58129F980A7957A85CC238D58A0A357D9FD6FCA1D7A8D713CBFAB4237708543"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	setReaderIp	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
signature	64	228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD	


```
{
	"apiVersion":"1.0",
	"message":"setReaderIp",
	"type":"ack",
	"timestamp":"2023-06-26 11:32:01",
	"messageTraceID":"112233445566",
	"body":{
		"errorCode":"0000"
	},
	"signature":"69A9CB53658F8ECB9F07C3ECED4EA33F7413B0ED0E1E9657317C89A14DA3A9BF"
}
```

4.13 Reboot

This function is to reboot the reader.

Request

Key Name	Max Length	Value
apiVersion	12	1.0
message	20	reboot
type	10	request
timestamp	20	2023-05-30 11:06:02
messageTraceID	40	112233445566
body	n	empty
signature	64	9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442


```
{
	"apiVersion":"1.0",
	"message":"reboot",
	"type":"request",
	"timestamp":"2023-06-26 11:30:30",
	"messageTraceID":"112233445566",
	"body":{},
	"signature":"5EE94E3BD613AAD3AA9355DAC44704F4654841F453BBF2F00BD969D27E54FE95"
}
```

<u>Response</u>

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	reboot	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
signature	64	228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD	


```
{
	"apiVersion":"1.0",
	"message":"reboot",
	"type":"ack",
	"timestamp":"2023-06-26 11:30:32",
	"messageTraceID":"112233445566",
	"body":{
		"errorCode":"0000"
	},
	"signature":"B1491B7B2D1557EB03AE8F85A147163E0A033E16EA7AAC7D2C54DFC49B1041ED"
}
```

4.14 Heartbeat

This function is an alternative way to maintain the heartbeat status.

Request

Key Name	Max Length	Value
apiVersion	12	1.0
message	20	ping
type	10	inform


```
{
        "apiVersion":"1.0",
        "message":"ping",
        "type":"inform"
}
```


Response

Key Name	Max Length	Value
apiVersion	12	1.0
message	10	ping
type	10	ack


```
{
        "apiVersion":"1.0",
        "message":"ping",
        "type":"ack"
}
```

4.15 Get Relay I/O Value

To obtain the I/O status from the I/O board connected to Jellies Reader.

Supported by ECPI-LPR firmware Version 6.13 and above (Beta release date 2025-OCT-29).

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	rioGet	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	port	0, 1, 2 or 3 which define the port number of I/O
signature	64	9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442	


```
{
        "apiVersion":"1.0",
        "message":"rioGet",
        "type":"request",
        "timestamp":"2023-06-26 11:30:30",
        "messageTraceID":"112233445566",
        "body":{
                "port":"0"
        },
        "signature":"E58129F980A7957A85CC238D58A0A357D9FD6FCA1D7A8D713CBFAB4237708543"
}
```


Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	rioGet	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	rioResp	0 - IO in open status 1 - IO in close status -1 - Invalid response from IO controller -2 - timeout from IO controller
		errorCode	0000 - Successful OTHER - Failed, Refer ERROR code table

signature	64	228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD	


```
{
	"apiVersion":"1.0",
	"message":"rioGet",
	"type":"ack",
	"timestamp":"2023-06-26 11:32:01",
	"messageTraceID":"112233445566",
	"body":{
		"errorCode":"0000"
		"rioResp":"1"
	},
	"signature":"69A9CB53658F8ECB9F07C3ECED4EA33F7413B0ED0E1E9657317C89A14DA3A9BF"
}
```

4.16 Set Relay I/O Value

To set the I/O to OPEN/CLOSE to the I/O board connected to Jellies Reader.

Supported by ECPI-LPR firmware Version 6.13 and above (Beta release date 2025-OCT-29)

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	rioSet	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	port	0, 1, 2 or 3 which define the port number of I/O
		ins	Instruction - 0 - Open - 1 - Close
signature	64	9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442	


```
{
	"apiVersion":"1.0",
	"message":"rioSet",
	"type":"request",
	"timestamp":"2023-06-26 11:30:30",
	"messageTraceID":"112233445566",
	"body":{
		"port":"0",
		“ins”:”1”
	},
	"signature":"E58129F980A7957A85CC238D58A0A357D9FD6FCA1D7A8D713CBFAB4237708543"
}
```

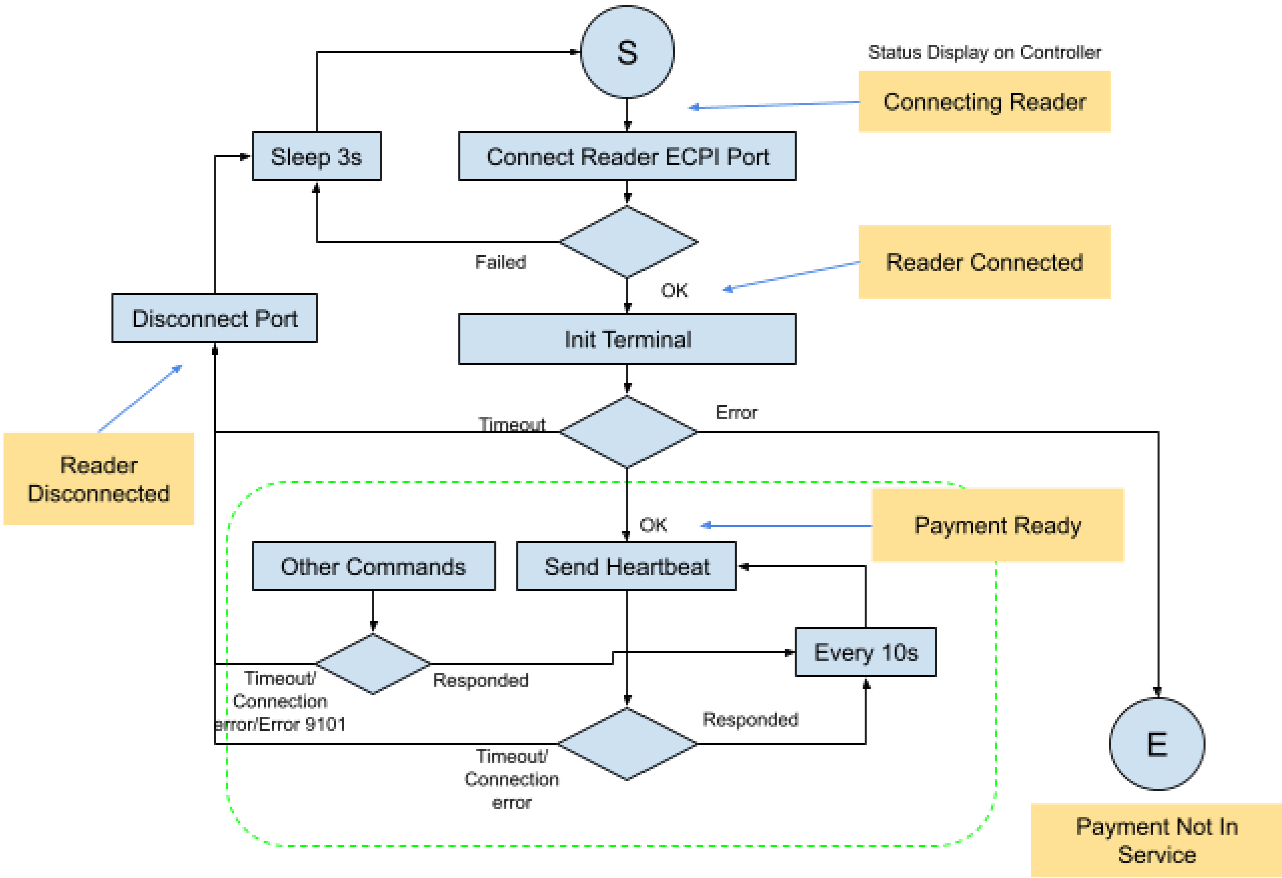

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	rioSet	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	rioResp	0 - IO OPEN/CLOSE successfully -1 - Invalid response from IO controller

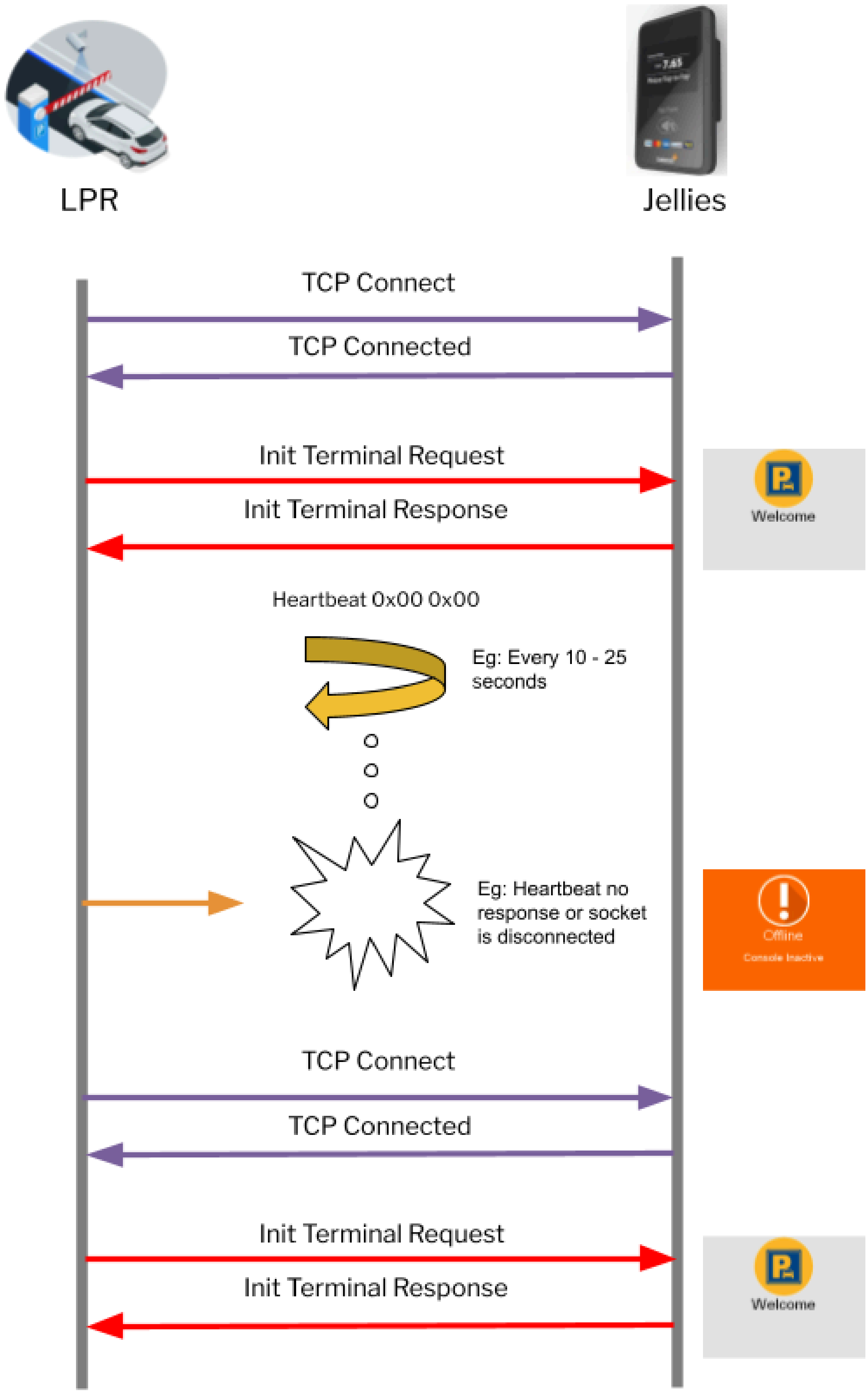
			-2 - timeout from IO controller
		errorCode	0000 - Successful OTHER - Failed, Refer ERROR code table
signature	64	228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD	


```
{
	"apiVersion":"1.0",
	"message":"rioGet",
	"type":"ack",
	"timestamp":"2023-06-26 11:32:01",
	"messageTraceID":"112233445566",
	"body":{
		"errorCode":"0000"
		"rioResp":"0"
	},
	"signature":"69A9CB53658F8ECB9F07C3ECED4EA33F7413B0ED0E1E9657317C89A14DA3A9BF"
}
```

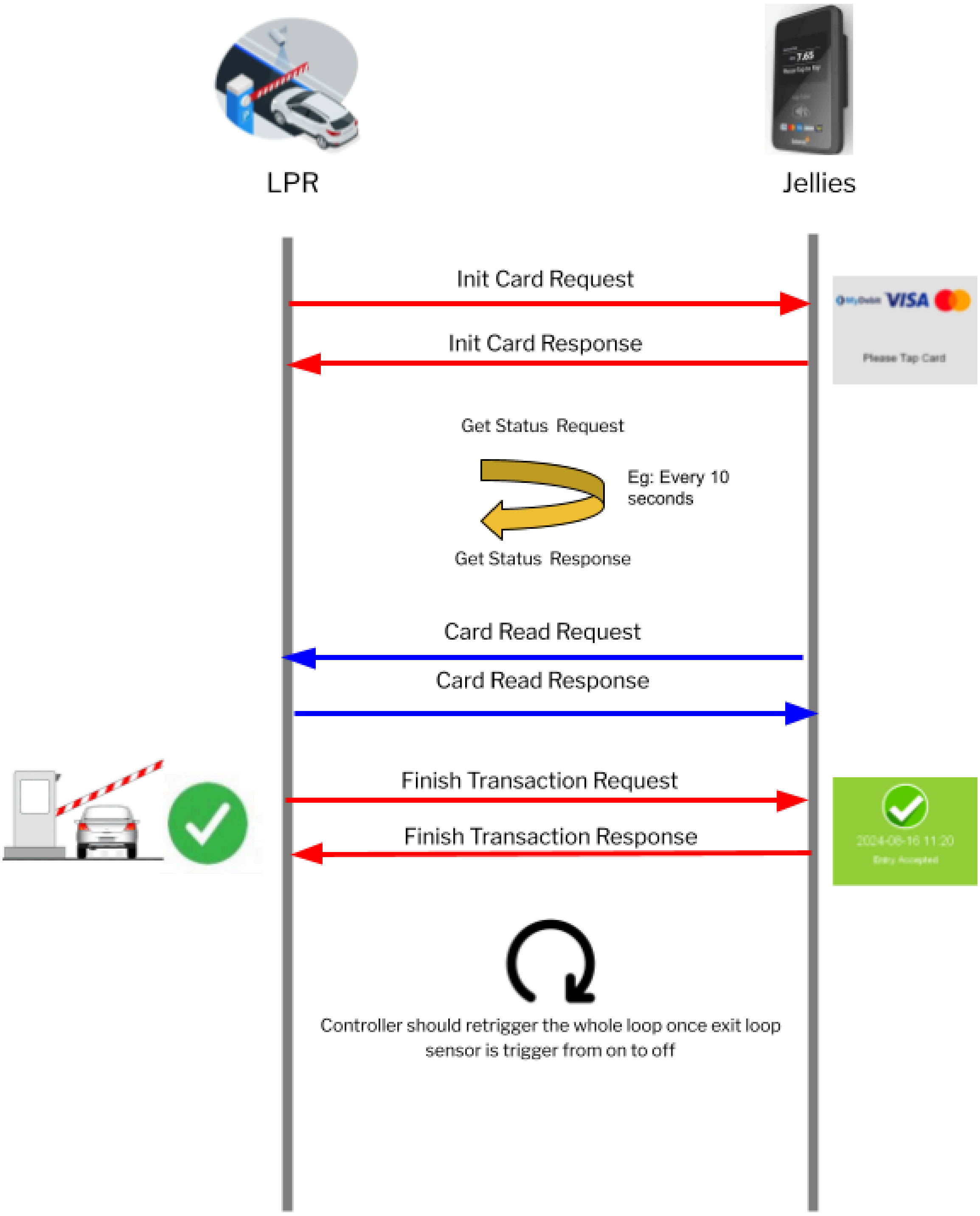
5 Process Flow



5.1 Initialization

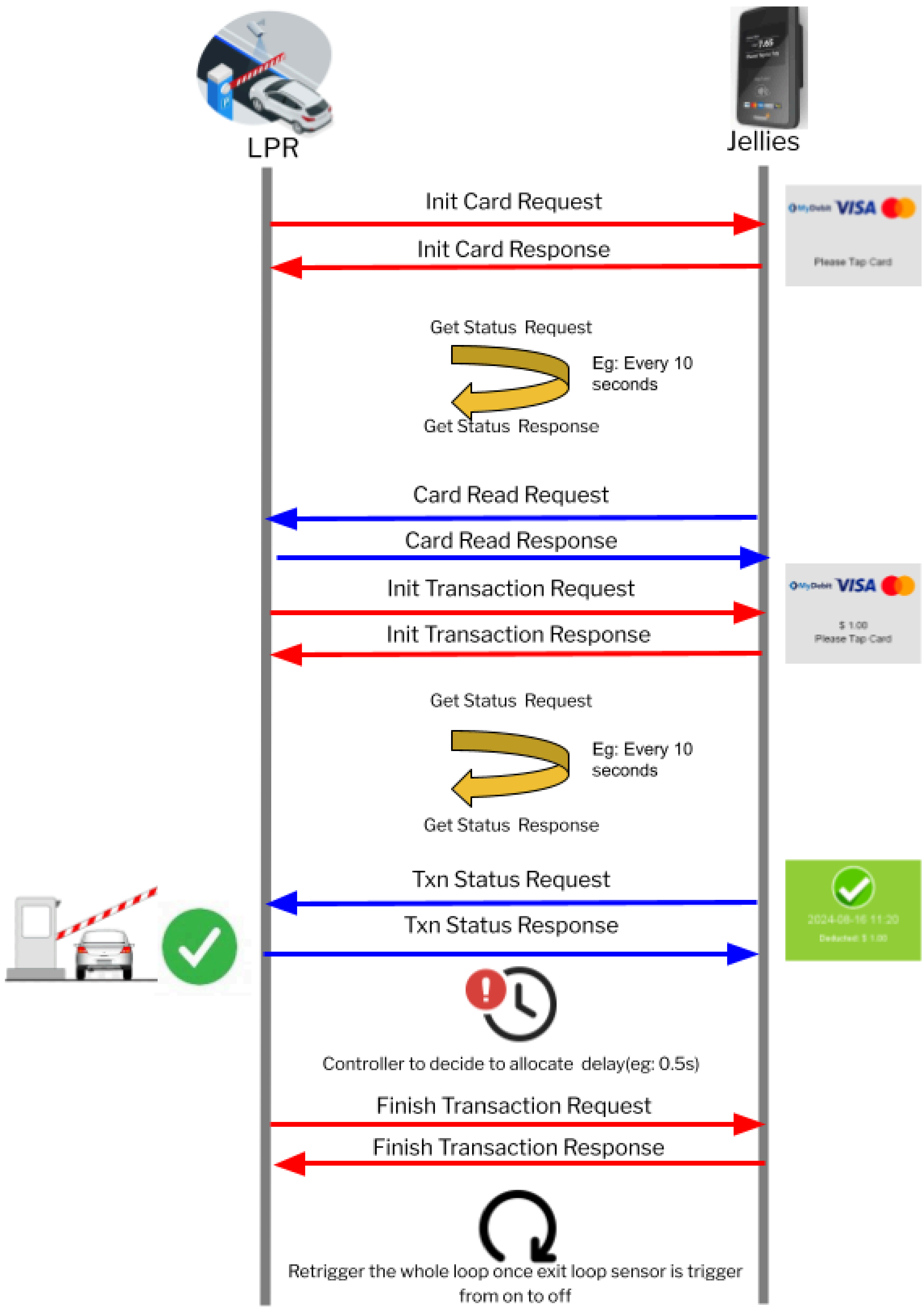


5.2 Entry

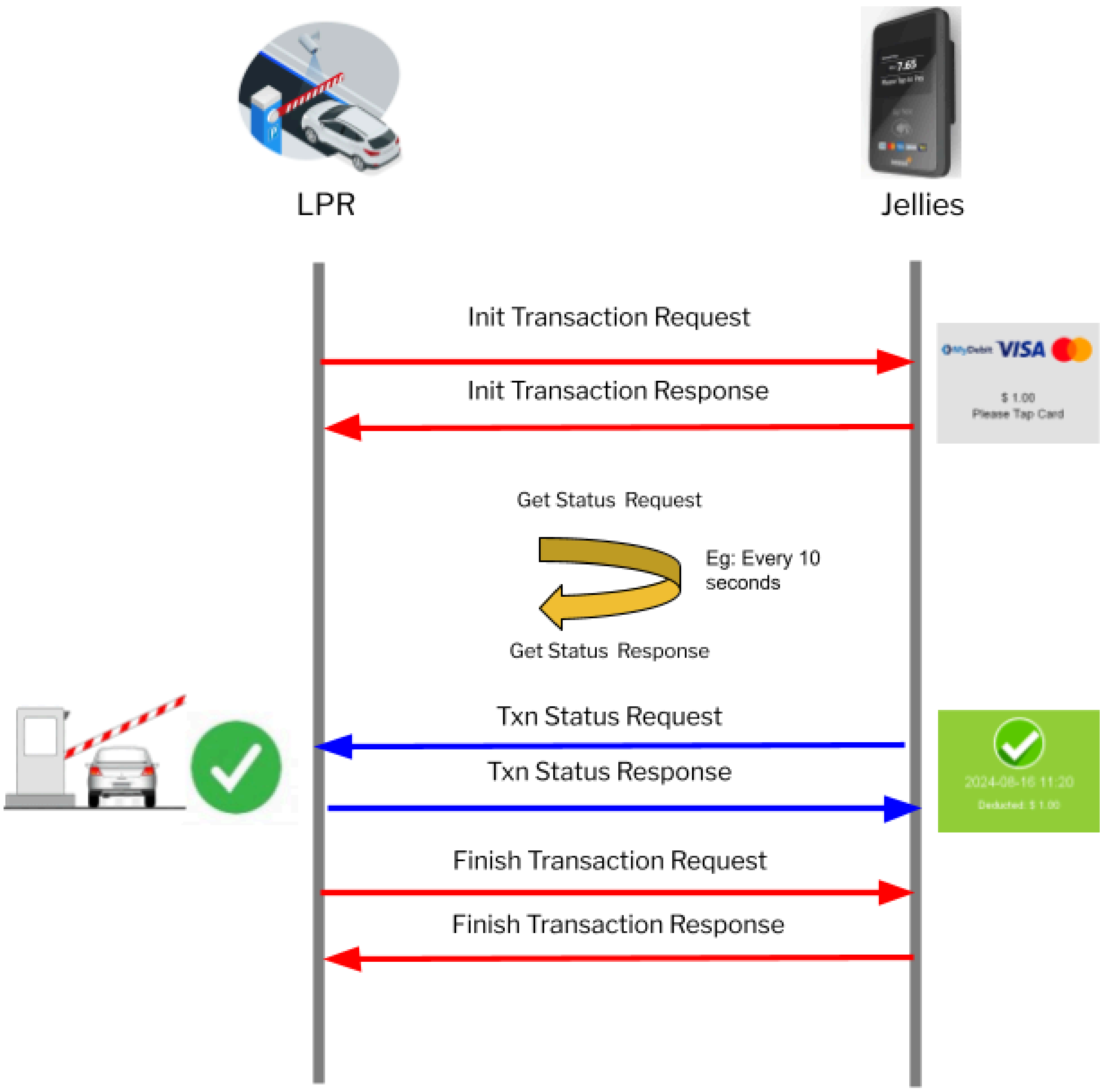

LPR
Jellies
Init Card Request
Init Card Response
Please Tap Card
Get Status Request
Eg: Every 10 seconds
Get Status Response
Card Read Request
Card Read Response
Finish Transaction Request
Finish Transaction Response
Controller should retrigger the whole loop once exit loop sensor is trigger from on to off

5.3 Exit

<u>Exit with card enter</u>

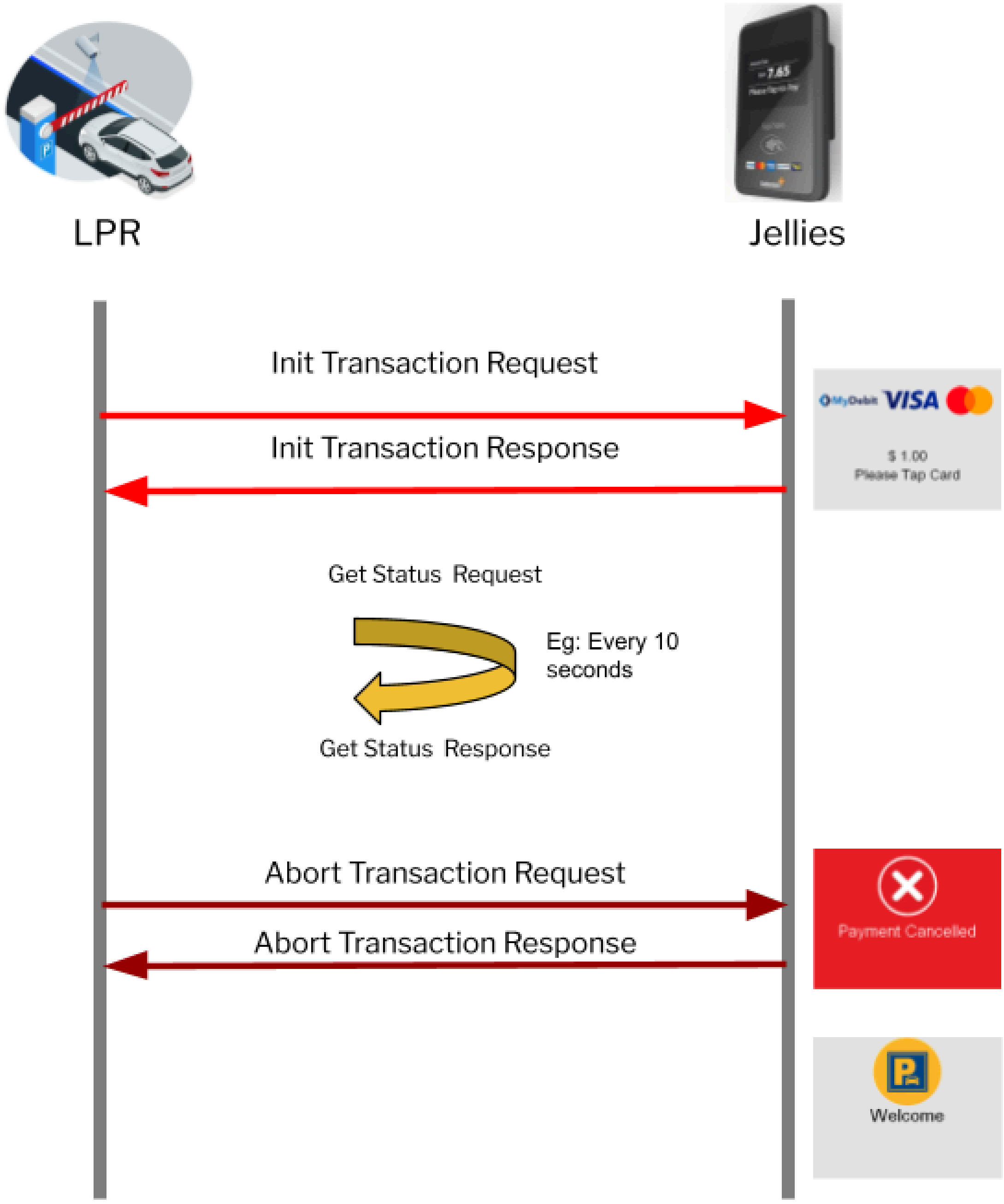


<u>Exit with LPR enter</u>



5.4 Abort Transaction

Car present to the loop and leave the loop without tapping card or payment successful



Appendix A

Error Code

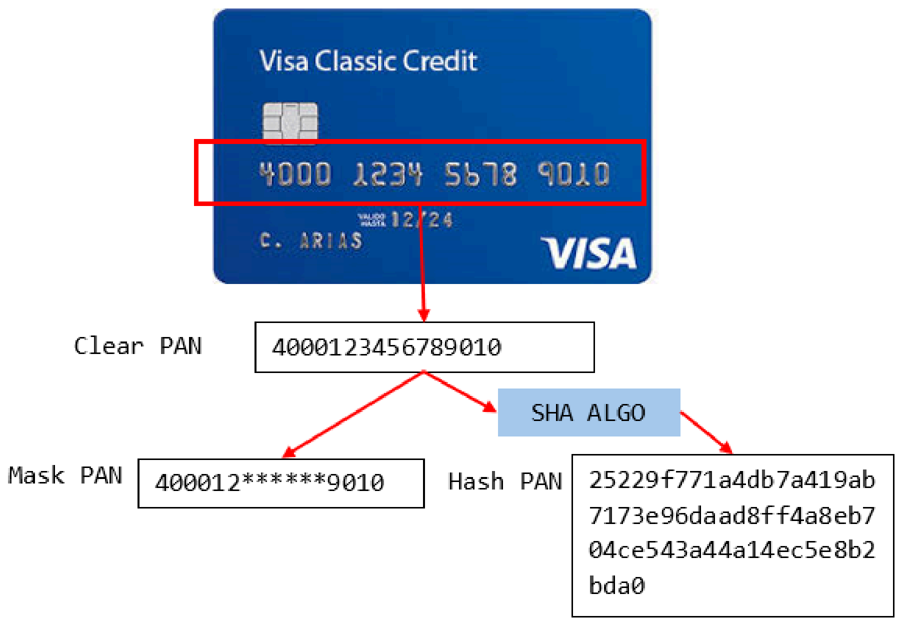
Error Code	Description
0000	Successful - Command executed successfully / Transaction Completed
10XX	Failed - Transaction Declined by Bank Host XX - Response Code from bank host 1005 - Do not honor 1051 - Insufficient Fund
1101	Failed - Transaction Failed
1102	Failed - Transaction Error
1199	Failed - Generic Error
2001	Failed - Invalid Data
3001	Failed - Card Invalid
3002	Failed - Card Blacklisted
3003	Failed - Card Not Allow
3004	Failed - Card Rejected
3005	Failed - Multiple Card
3006	Failed - Card Expired
3007	Failed - Insufficient Fund
3101	Failed - Bank Connect Error
3102	Failed - Bank Timeout
3201	Failed - Aborted by Kiosk
9001	Failed - Terminal Busy
9101	Failed - Terminal Not Active **Controller has to resend initTerminal in order to activate the reader**
9102	Failed - Terminal Active
9999	Failed - General Error

Reader State

State	Description
1	Idle
2	Card Scanning
3	Card Detected
4	Host Authorizing
5	Completion

Appendix B

Mask Pan and Hash Pan



Example

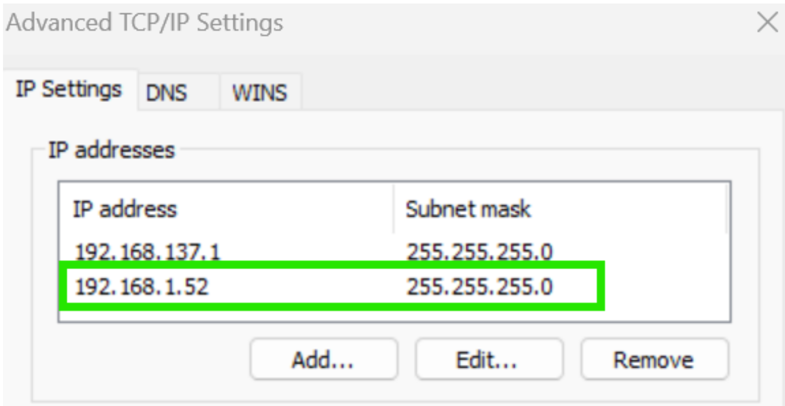
Clear PAN	Mask PAN	Hash PAN (20bytes, 40chars)
4000123456789010	400012******9010	25229f771a4db7a419ab7173e9 6daad8ff4a8eb704ce543a44a1 4ec5e8b2bda0
4000127654329010	400012******9010	3521122e5308163881cd1d31dff 6be11f6b8dbc9fee8083d93a523 6761f85e62

Use Mask PAN for report – Human Recognizable.
Use Hash PAN for system matching. Prevent duplicable Mask PAN.

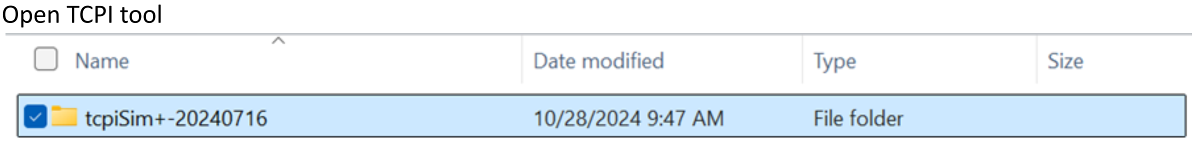
Appendix C

Set reader IP using TCPI tool

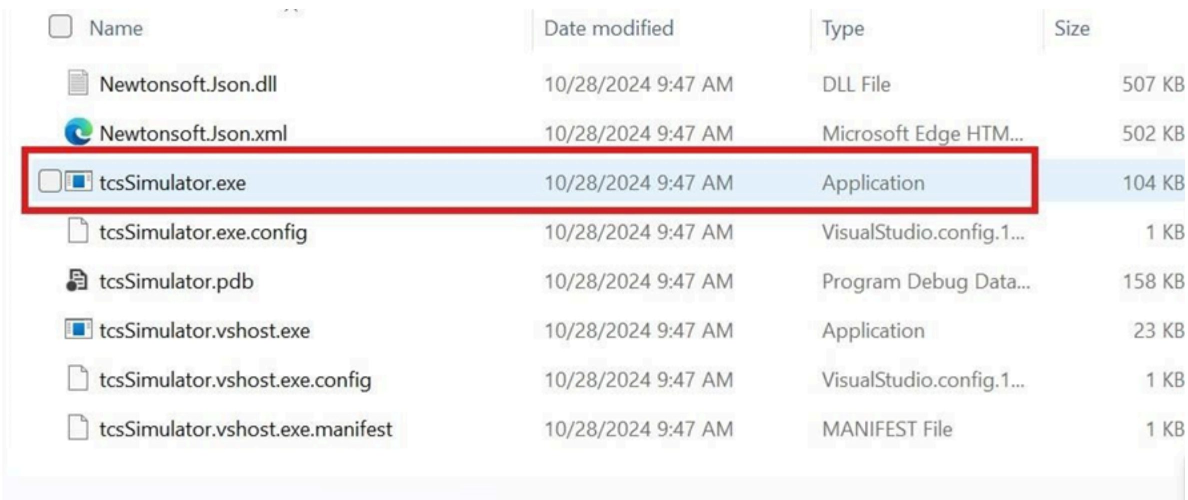
1. Ensure to configure an additional IP at IPV4 of same subnet with the reader 192.168.1.x. (Eg: 192.168.1.52) and ping to test the connection with reader.



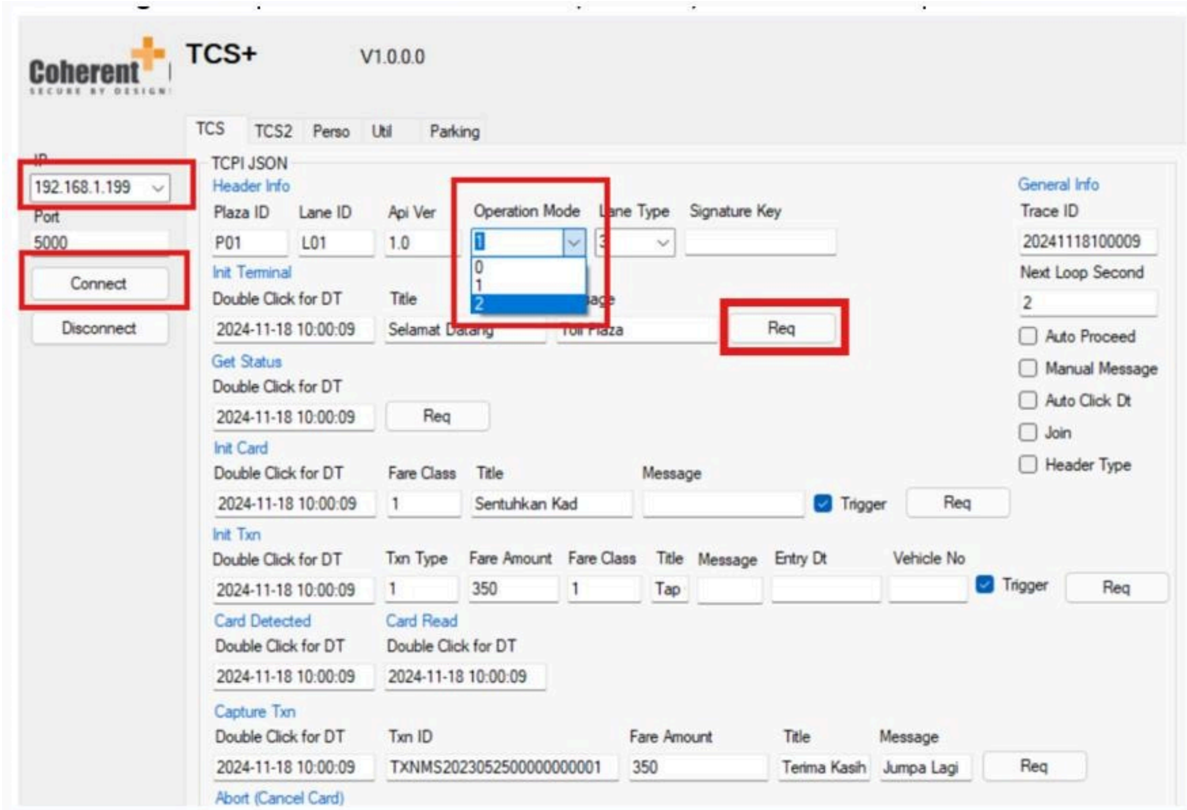
2. Open TCPI tool



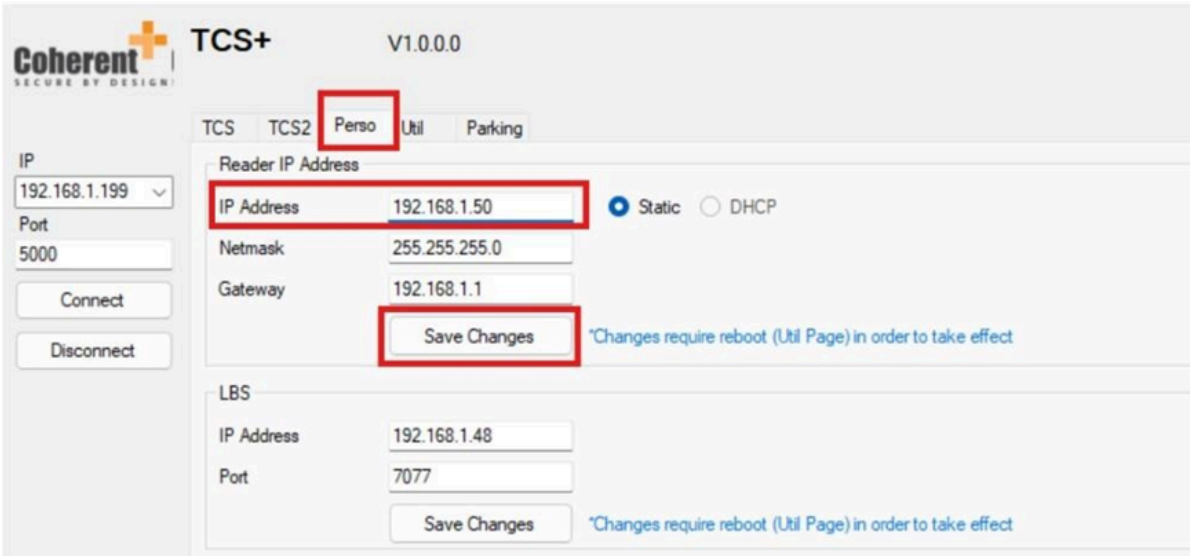
3. Launch TCPI



4. Enter the current reader's IP address, and click "connect". Once connected, change the operator mode to "2" (offline mode) and click "Req".

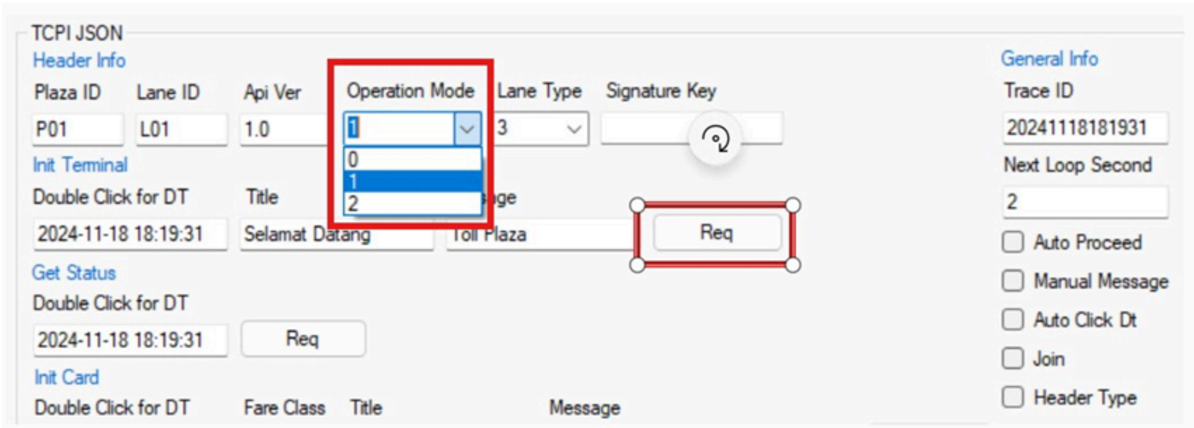


5. Select "perso" tab and type in new reader IP and "save changes".



6. Go to “Util” tab and click “reboot”.

7. Init terminal in operation mode “1” (Live mode).



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